

Integration WS 6

Evaluate Q1-15

1. $\int \frac{1}{1+e^{-x}} dx$
2. $\int \ln(x^2 - 1) dx$
3. $\int \frac{1+\sin x}{\cos^2 x} dx$
4. $\int \sin 5x \cos 3x dx$
5. $\int \frac{1}{3+2x-x^2} dx$
6. $\int \frac{1}{e^x + e^{-x}} dx$
7. $\int \ln(x^2 + 1) dx$
8. $\int \frac{1}{\sqrt{3+2x-x^2}} dx$
9. $\int \frac{x^2}{x^2-1} dx$
10. $\int \frac{dx}{x^2\sqrt{x^2-4}}$
11. $\int \sqrt{4-x^2} dx$
12. $\int \frac{1}{x} \sec^2(\ln x) dx$
13. $\int \frac{2x+1}{x^2+2x+2} dx$
14. $\int \sin^3 x \cos^2 x dx$
15. $\int \sqrt{16+x^2} dx$

16. For $n = 0, 1, 2, \dots$ let $I_n = \int_0^{\frac{\pi}{4}} \tan^n \theta d\theta$

- a. Show that $I_1 = \frac{1}{2} \ln 2$
- b. Show that, for $n \geq 2$, $I_n + I_{n-2} = \frac{1}{n-1}$
- c. For $n \geq 2$, explain why $I_n < I_{n-2}$, and deduce that $\frac{1}{2(n+1)} < I_n < \frac{1}{2(n-1)}$
- d. Use the reduction formula in Part (b) to find I_5 , and hence deduce that $\frac{2}{3} < \ln 2 < \frac{3}{4}$.

17. Find:

- a. $\int x^3 \ln x dx$
- b. $\int \sin^3 2x dx$
- c. $\int \frac{1}{x^2+6x+25} dx$
- d. $\int 3x \cos 3x dx$

e. $\int \frac{x}{\sqrt{4+x}} dx$

18. Let $T_n = \int_0^1 x^n \sqrt{1-x} dx$

a. Deduce the reduction formula $T_n =$

$$\frac{2n}{2n+3} T_{n-1}$$

b. Show that $T_3 = \frac{32}{315}$

c. Use the reduction formula to help prove by

induction that $T_n = \frac{n!(n+1)!}{(2n+3)!} 4^{n+1}$

19. Consider the integral $I_n = \int_0^1 \frac{x^n}{\sqrt{1+x}} dx$

a. Show that $I_0 = 2\sqrt{2} - 2$

b. Show that $I_{n-1} + I_n = \int_0^1 x^{n-1} \sqrt{1+x} dx$

c. Use integration by parts to show that

$$I_n = \frac{2\sqrt{2}-2nI_{n-1}}{2n+1} \text{ for } n \geq 1.$$

d. Hence evaluate I_2

20. Let $I_n = \int_0^1 (1-x^2)^n dx$ and $J_n = \int_0^1 x^2(1-x^2)^n dx$

a. Apply integration by parts to I_n to show that $I_n = 2nJ_{n-1}$

b. Hence show that $I_n = \frac{2n}{2n+1} I_{n-1}$ for $n \geq 1$

c. Show that $J_n = I_n - I_{n+1}$, and hence deduce that $J_n = \frac{1}{2n+3} I_n$

d. Hence write down a reduction formula for J_n in terms of J_{n-1}

21. Determine:

a. $\int \frac{1}{1+\cos x} dx$

b. $\int \frac{1}{1+\sin x - \cos x} dx$

c. $\int \frac{1}{3 \sin x + 4 \cos x} dx$

22. a. Use the t-substitution to show that

$$\int \sec x dx = \log \left(\frac{1+\tan \frac{x}{2}}{1-\tan \frac{x}{2}} \right) + C$$

- b. Show that this answer and $\int \sec x \, dx = \ln(\sec x + \tan x) + C$ is equivalent.
23. Use integration by parts to find the following. You may also need to apply the Pythagorean identity $1 + \tan^2 x = \sec^2 x$.
- $\int \sec x \tan^2 x \, dx$
 - $\int \sec^3 x \, dx$
 - $\int 8 \sec^5 x \, dx$
24. $\int 8 \sec^5 x \, dx$ Let $I_n = \int_1^e (1 - \ln x)^n \, dx$ for $n = 1, 2, 3 \dots$
- Show that $I_n = -1 + nI_{n-1}$
 - Hence evaluate $\int_1^e (1 - \ln x)^3 \, dx$
 - Show that $\frac{I_n}{n!} = e - \sum_{r=0}^n \frac{1}{r!}$
25. Using the substitution $t = \tan \frac{\theta}{2}$ to find $\int_0^{\frac{\pi}{2}} \frac{d\theta}{3 \sin \theta + 4 \cos \theta + 5}$
26. a. If $I_n = \int \operatorname{cosec}^n x \, dx$ for $n \geq 0$ show that
- $$I_n = \frac{-\cot x \operatorname{cosec}^{n-2} x}{n-1} + \frac{n-2}{n-1} I_{n-2} \text{ given}$$
- $$\frac{d}{dx}(\cot x) = -\operatorname{cosec}^2 x.$$
- b. Hence show that if $I_n = \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \operatorname{cosec}^n x \, dx$ for $n \geq 0$, then $I_n = \frac{(\sqrt{2})^{n-2}}{n-1} + \frac{n-2}{n-1} I_{n-2}$, for $n \geq 2$
- c. Deduce that $I_6 = \frac{28}{15}$
27. a. Let $I_n = \int_0^1 x^n \sqrt{1-x^3} \, dx$ for $n \geq 2$.
- Show that $I_n = \frac{2n-4}{2n+5} I_{n-3}$ for $n \geq 5$.
 - Hence find I_8 .
28. a. Let $u_n = \int_0^{\frac{\pi}{2}} \sin^n x \cos^2 x \, dx$.
- Show that $u_n = \frac{n-1}{n+2} u_{n-2}$, for $n \geq 2$.
 - Hence show that $u_4 = \frac{\pi}{32}$
29. a. If $f(x) = f(a-x)$, prove that $\int_0^a x f(x) \, dx = \frac{a}{2} \int_0^a f(x) \, dx$
- b. Hence, or otherwise, prove that $\int_0^{\pi} g(x) \, dx = \frac{\pi^2}{4}$, if $g(x) = \frac{x \sin x}{1 + \cos^2 x}$
30. a. Let $I_n = \int_0^{\frac{\pi}{2}} (\sin x)^n \, dx$, where n is an integer, $n \geq 0$
- Using integration by parts, show that for $n \geq 2$, $I_n = \left(\frac{n-1}{n}\right) I_{n-2}$
 - Deduce that $I_{2n} = \frac{2n-1}{2n} * \frac{2n-3}{2n-2} * \dots * \frac{3}{4} * \frac{1}{2} * \frac{\pi}{2}$ and $I_{2n+1} = \frac{2n}{2n+1} * \frac{2n-2}{2n-1} * \dots * \frac{4}{5} * \frac{2}{3} * 1$
 - Explain why $I_k > I_{k+1}$
 - Hence, using the fact that $I_{2n-1} > I_{2n}$ and $I_{2n} > I_{2n+1}$, show that $\frac{\pi}{2} \left(\frac{2n}{2n+1}\right) < \frac{2^2 * 4^2 * \dots * (2n)^2}{1 * 3^2 * 5^2 * \dots * (2n-1)^2 (2n+1)} < \frac{\pi}{2}$

Answer

- $\ln(e^x + 1)$
- $x \ln|x^2 - 1| + \ln \left| \frac{x+1}{x-1} \right| - 2x$
- $\tan x + \sec x$
- $-\frac{1}{16} \cos 8x - \frac{1}{4} \cos 2x$
- $\frac{1}{4} \ln \left| \frac{1+x}{3-x} \right|$
- $\tan^{-1} e^x$
- $x \ln(x^2 + 1) + 2 \tan^{-1} x - 2x$
- $\sin^{-1} \left(\frac{x-1}{2} \right)$
- $x - \frac{1}{2} \ln \left| \frac{x+1}{x-1} \right|$
- $\frac{1}{4x} \sqrt{4-x^2}$
- $\frac{1}{2} x \sqrt{4-x^2} + 2 \sin^{-1} \frac{x}{2}$
- $\tan(\ln x)$
- $\ln(x^2 + 2x + 2) - \tan^{-1}(x + 1)$
- $-\frac{1}{3} \cos^3 x + \frac{1}{5} \cos^5 x$
- $\frac{1}{2} x \sqrt{16+x^2} + 8 \ln\{x + \sqrt{16+x^2}\}$
- $d. I_5 = \frac{1}{4} (2 \ln 2 - 1)$
- a. $\frac{1}{4} x^4 \ln x - \frac{1}{16} x^4 + C$; b. $\frac{1}{6} \cos^3 2x - \frac{1}{2} \cos 2x + C$; c. $\frac{1}{4} \tan^{-1} \frac{x+3}{4} + C$; d. $x \sin 3x + \frac{1}{3} \cos 3x + C$; e. $\frac{2}{3} (x-8) \sqrt{4+x} + C$
- prove
- d. $\frac{1}{15} (14\sqrt{2} - 16)$
- d. $J_n = \frac{2n}{2n+3} J_{n-1}$
- a. $\tan \frac{x}{2} + C$; b. $\log \left(\frac{\tan \frac{x}{2}}{1 + \tan \frac{x}{2}} \right) + C$; c. $\frac{1}{5} \log \left(\frac{1 + 2 \tan \frac{x}{2}}{2 - \tan \frac{x}{2}} \right) + C$
- prove
- a. $\frac{1}{2} (\sec x \tan x - \ln(\sec x + \tan x)) + C$; b. $\frac{1}{2} (\sec x \tan x + \ln(\sec x + \tan x)) + C$; c. $\sec x \tan x (2 \sec^2 x + 3) + 3 \ln(\sec x + \tan x) + C$
- $6e - 16$
- $\frac{1}{6}$
- $\frac{16}{315}$